THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE FULL SYLLABUS PAPER SET 06

Full Syllabus (Chapters 1–13) | RBSE Board Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. A balanced chemical equation is based on the law of: 1

(a) multiple proportions **(b)** conservation of energy
(c) constant proportions **(d)** conservation of mass

2. Student A says baking soda is $NaHCO_3$; Student B says it is Na_2CO_3. Then: 1

(a) Student A is correct **(b)** Student B is correct
(c) Both are correct **(d)** Both are wrong

3. The number of water molecules in gypsum ($CaSO_4 \cdot x\,H_2O$) is: 1

(a) 10 **(b)** 5 **(c)** ½ **(d)** 2

4. Bauxite is an ore of: 1

(a) zinc **(b)** iron **(c)** aluminium **(d)** copper

5. Which gas is evolved when a metal reacts with dilute HCl? 1

(a) oxygen **(b)** carbon dioxide **(c)** chlorine **(d)** hydrogen

6. A student states: “Ethanol is a good conductor of electricity.” The correct response is: 1

(a) Correct **(b)** Incorrect — it does not ionise
(c) Correct when warm **(d)** Cannot be determined

7. Carbon forms the largest number of compounds because of: 1

(a) metallic nature **(b)** catenation and tetravalency
(c) ionic bonding **(d)** high reactivity

8. In single-celled organisms, gas exchange occurs by: 1

(a) gills **(b)** diffusion **(c)** lungs **(d)** tracheae

9. The brain and spinal cord together form the: 1

(a) central nervous system **(b)** autonomic system
(c) reflex arc **(d)** peripheral nervous system

10. Leishmania causes: 1

(a) AIDS **(b)** malaria **(c)** kala-azar **(d)** typhoid

11. The plant in which reproduction by fragmentation is common is: 1

(a) potato **(b)** rose **(c)** Spirogyra **(d)** mango

12. The probability that a child of XX × XY parents will be female is: 1

(a) 1/2 **(b)** 1/4 **(c)** 1 **(d)** 3/4

13. The gene for a hormone-producing enzyme being less efficient results in: 1

(a) no change **(b)** taller plants **(c)** shorter plants **(d)** white flowers

14. A spherical mirror and a thin spherical lens each have focal length −15 cm. They are: 1

(a) both concave mirror and concave lens **(b)** both convex

(c) convex mirror, concave lens **(d)** concave mirror, convex lens

15. The apparent flattening of the Sun's disc at sunrise is due to: 1

(a) scattering **(b)** reflection

(c) dispersion **(d)** atmospheric refraction

16. The potential difference between two points is measured in: 1

(a) volt **(b)** watt **(c)** ohm **(d)** ampere

17. A wire of resistance R is cut into two equal halves, which are joined in parallel. The resistance becomes: 1

(a) $2R$ **(b)** R **(c)** $R/2$ **(d)** $R/4$

18. The rating of a fuse used for a 2 kW geyser at 220 V is about: 1

(a) 50 A **(b)** 10 A **(c)** 1 A **(d)** 5 A

19. The harmful UV rays are absorbed by: 1

(a) carbon dioxide layer **(b)** ozone layer

(c) nitrogen **(d)** clouds

20. A student states: “Plastic can be broken down by bacteria quickly.” The correct response is: 1

(a) Correct **(b)** Correct in water

(c) Incorrect — it is non-biodegradable **(d)** Cannot be determined

SECTION — B (Very Short Answer Type) 8 × 2 = 16

Answer in about 10–15 words each.

21. What is a skeletal chemical equation? Give an example. 2

22. What is the action of dilute HCl on copper oxide? Write the equation. 2

23. Why is it necessary to separate oxygenated and deoxygenated blood in mammals and birds? 2

24. Name the hormone released by the thyroid and its deficiency disease. 2

25. Why does menstruation occur? 2

26. Name the scientist known as the father of genetics and the plant he used. 2

27. Light travels from air to glass ($n = 1.5$). If the speed in air is 3×10^8 m/s, find the speed in glass. 2

28. Why does a current-carrying conductor experience a force in a magnetic field? 2

SECTION — C (Short Answer Type) 8 × 3 = 24

Answer in about 30–40 words each.

29. A grey metal M reacts with steam to form a black oxide and a gas that burns with a pop sound. Identify M and the oxide and write the equation. 3

30. Why is iron galvanised? What happens if the zinc coating is scratched? 3

31. How is the brain protected? Why is it necessary? 3

32. What is budding? Explain with Hydra and a diagram. 3

33. Why is the colour of the clear sky blue? Explain with the size of particles. 3

34. Why does a compass placed near a current-carrying wire deflect? What happens when the current is switched off? 3

35. Why is pH important for our digestive system? What happens during indigestion and how is it treated? 3

OR

Give the chemical name, formula and one use of 3
(i) washing soda
(ii) baking soda
(iii) bleaching powder.

36. How do decomposers help in recycling nutrients? Give two examples. 3

OR

Give one example each of a terrestrial food chain with 3
(i) three
(ii) four trophic levels.

SECTION — D (Long Answer Type) 4 × 5 = 20

Answer in about 100–150 words each.

37. (i) Derive the meaning of magnification for a spherical mirror. 5
(ii) An object is placed 6 cm in front of a concave mirror of focal length 12 cm. Find the position and nature of the image.

OR

(i) Define power of a lens and its SI unit. 5
(ii) Find the focal length of a lens of power −2.0 D. What type of lens is it?
(iii) A convex lens of +5 D and a concave lens of −3 D are in contact. Find the net power and focal length.

38. (i) What is a homologous series? 5
(ii) Write the first three members of the alcohol series with formulae.
(iii) How do their physical and chemical properties vary?

OR

(i) Explain the nomenclature of carbon compounds with the help of prefixes/suffixes for halogen, alcohol, aldehyde and ketone. 5
(ii) Name: CH_3CH_2Br, CH_3CH_2CHO.

39. (i) State Ohm's law and its limitations. 5
(ii) Plot the V–I graph from the data V (V): 1.6, 3.4, 6.7, 10.2, 13.2; I (A): 0.1, 0.2, 0.4, 0.6, 0.8 (describe). Find R.

OR

(i) Explain electric power and its commercial unit. 5
(ii) A 1200 W iron is used 1 h daily and a 60 W bulb 10 h daily. Find the monthly bill (30 days) at ₹5 per unit.

40. (i) How does digestion of carbohydrates, proteins and fats take place in humans? Name the enzymes. 5
(ii) Where is the digested food absorbed?

OR

(i) Describe the passage of air from nostrils to alveoli. 5

(ii) How are alveoli adapted for gas exchange?

(iii) Why does the breathing rate increase during exercise?

END OF QUESTION PAPER **Total: 80 Marks**

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